

Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 10: Respiration in Organisms

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	10 — Respiration in Organisms
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Traps target the classic confusions: equating breathing with respiration, thinking plants don't respire by day, believing anaerobic respiration releases as much energy as aerobic, assuming exhaled air has zero oxygen, confusing the organs of different animals (gills/spiracles/tracheae/skin/stomata/lenticels), reversing the diaphragm/rib movements, mixing up the pressure logic of inhalation/exhalation, and reading breathing-rate data backwards. Do not write on this paper — mark answers on your answer sheet.

1. A teacher writes 'glucose + oxygen → carbon dioxide + water + energy' on the board and asks which item on the right-hand side is NOT a substance you could collect in a jar. The best answer is:

- (A) energy, because it is released, not a material substance
- (B) water, because water is not produced in respiration
- (C) carbon dioxide, because it is used up rather than made
- (D) oxygen, because oxygen appears on the right side too

2. A student claims, 'When I run, I breathe faster, so I respire faster.' The most accurate response is:

- (A) breathing and respiring are unrelated, so faster breathing changes nothing inside the cells
- (B) breathing faster slows respiration because the lungs cannot keep up
- (C) respiration only happens in the lungs, so cell respiration does not change
- (D) faster breathing supplies the extra oxygen the cells need to respire faster, so both speed up together

3. Per molecule of glucose, aerobic respiration releases far more energy than anaerobic respiration. The best reason is that:

- (A) anaerobic respiration uses extra oxygen, which wastes energy
- (B) aerobic respiration produces alcohol, which stores most of the energy
- (C) anaerobic respiration happens faster, so its energy escapes as heat
- (D) with oxygen the glucose is broken down completely into carbon dioxide and water, whereas without oxygen it is only partly broken down

4. Yeast added to warm sugar dough makes the dough rise. Connecting cause to effect, the rising is caused by:

- (A) bubbles of carbon dioxide released by yeast as it respire the sugar without enough oxygen
- (B) oxygen given off by the yeast as it photosynthesises the sugar
- (C) water vapour from the yeast boiling the dough
- (D) lactic acid produced by the yeast making the dough swell

5. Two glasses of grape juice are left out: one with baker's yeast and one without. After a day the yeast glass smells of alcohol and bubbles, while the plain glass does not. This shows that:

- (A) the yeast turned the alcohol back into sugar and oxygen
- (B) the yeast photosynthesised the sugar into alcohol
- (C) the plain juice respired faster than the yeast juice
- (D) the yeast carried out anaerobic respiration of the sugar, producing alcohol and carbon dioxide

6. After a 200 m sprint a runner's calf cramps, but it eases after she stops and rests. The cramp eased mainly because:

- (A) resting made the muscle produce more lactic acid to relax it
- (B) resting cooled the alcohol that had formed in the muscle
- (C) resting let oxygen reach the muscle and break down the lactic acid that had built up
- (D) resting refilled the muscle with carbon dioxide

7. Which everyday situation best matches anaerobic respiration in human muscles?

- (A) sitting quietly and reading a book (B) taking a slow gentle walk in the garden
- (C) lifting a very heavy load suddenly, so the muscles work faster than oxygen can be supplied
- (D) sleeping deeply through the night

8. A child says, 'Lungs are where energy is made in the body.' The correct correction is:

- (A) lungs only exchange gases; energy is released by respiration inside every cell of the body
- (B) lungs make energy, but only when we exercise

(C) energy is made in the nose, not the lungs

(D) energy is made in the blood as it passes through the lungs

9. At rest a boy breathes 16 times a minute; while jogging he breathes 28 times a minute. How many MORE breaths does he take per minute while jogging?

(A) 28

(B) 44

(C) 12

(D) 16

10. A man breathes 15 times per minute at rest. If he keeps this rate, about how many breaths does he take in half an hour?

(A) 45

(B) 450

(C) 900

(D) 30

11. A table shows breathing rates: sleeping 12, standing 16, walking 20, running 36 (breaths per minute). The pattern best supports the idea that:

(A) breathing rate falls as activity increases

(B) the harder the muscles work, the faster the body must breathe to supply oxygen

(C) breathing rate stays the same whatever the activity

(D) the body breathes fastest when completely at rest

12. Two students, A and B, are the same age. After identical exercise A breathes 22/min and B breathes 30/min, and A is the trained athlete. The most reasonable reading is that:

(A) A's fitter body delivers oxygen more efficiently, so A needs fewer breaths for the same work

(B) B is fitter because a higher breathing rate means stronger lungs

(C) the two readings show respiration has stopped in A

(D) A breathes less because A's cells do not need oxygen

13. Climbers on a high mountain breathe faster even while standing still. The best Class-7 explanation is that:

(A) the cold air forces the diaphragm to move faster

(B) high altitude air is full of carbon dioxide that must be cleared quickly

(C) the lungs shrink at height and can hold less air each time

(D) the air is thinner there, so each breath carries less oxygen and more breaths are needed

14. Trace one oxygen molecule from outside to the blood. The correct ordered path is:

(A) nostrils → trachea → nasal cavity → alveoli → bronchi → blood

(B) nostrils → nasal cavity → trachea → bronchi → alveoli → blood

(C) nasal cavity → nostrils → bronchi → trachea → alveoli → blood

(D) nostrils → nasal cavity → bronchi → trachea → alveoli → blood

15. During a normal inhalation, the diaphragm and the ribs do what, and with what effect on chest volume?

- (A) diaphragm domes up while ribs move down, so chest volume increases
- (B) diaphragm flattens down while ribs move down and in, so chest volume decreases
- (C) diaphragm flattens and moves down while ribs move up and out, so chest volume increases
- (D) diaphragm domes up while ribs move up, so chest volume decreases

16. Why does air actually rush INTO the lungs during inhalation?

- (A) the lungs actively suck air in like a vacuum cleaner motor
- (B) the chest expands, so the pressure inside the lungs falls below the outside pressure and air flows in
- (C) the diaphragm pushes air up into the lungs from the stomach
- (D) warm air outside naturally rises into the cooler lungs

17. During exhalation the air leaves the lungs because:

- (A) the chest expands and the pressure inside the lungs drops
- (B) the lungs make extra carbon dioxide that pushes the air out
- (C) the diaphragm relaxes and the chest shrinks, raising the pressure inside the lungs above the outside
- (D) the ribs swing further up and out, squeezing the lungs

18. A model lung is made from a bottle with a rubber sheet across the bottom and a balloon inside. When you pull the rubber sheet DOWN, the balloon inflates. The rubber sheet represents the ____ and pulling it down models ____.

- (A) diaphragm; exhalation
- (B) ribs; exhalation
- (C) lungs; breathing out
- (D) diaphragm; inhalation

19. A cockroach is sealed in a jar; tiny openings along its body let air in. These openings and the tubes that carry the air directly to its tissues are, respectively:

- (A) gills and bronchi
- (B) stomata and lenticels
- (C) spiracles and tracheae
- (D) nostrils and alveoli

20. A fish moved from river water into a sealed bag of water that has been boiled and cooled soon becomes distressed. The best reason is that:

- (A) boiled water has too much oxygen, which poisons the gills
- (B) the fish needs carbon dioxide dissolved in the water to breathe
- (C) boiled water is too pure for the fish to swallow
- (D) boiling drove out the dissolved oxygen, and the gills have none to extract

21. Why must a fish keep pumping water over its gills almost constantly, while a human breathes air far less often for the same activity?

- (A) water holds much less dissolved oxygen than air, so far more water must pass the gills
- (B) fish respire many times faster than humans do

- (C) gills can only take oxygen from moving water, not still water
- (D) water is heavier, so it carries oxygen away before the gills can use it

22. After heavy rain, earthworms crawl out onto the soil surface. The best respiration-based reason is:

- (A) earthworms come up to drink the rainwater they need to breathe
- (B) the worms switch to breathing through gills at the surface
- (C) wet soil has extra oxygen that the worms come up to avoid
- (D) rainwater fills the air spaces in the soil, so the worm's moist skin cannot get enough oxygen underground

23. A frog can stay underwater for a long time without surfacing. Which statement best explains this?

- (A) it has gills like a fish that work underwater
- (B) it stores enough oxygen in its stomach for hours
- (C) it stops respiring completely while submerged
- (D) it can take in oxygen through its moist skin as well as through its lungs

24. Match each organism to the structure it uses to take in oxygen. Which set is entirely correct?

- (A) fish — spiracles; cockroach — gills; earthworm — lungs
- (B) fish — gills; cockroach — spiracles; earthworm — moist skin
- (C) fish — moist skin; cockroach — stomata; earthworm — gills
- (D) fish — lungs; cockroach — tracheae only sealed off; earthworm — spiracles

25. A green leaf is sealed in a clear box at midday in bright sunlight. Which is the MOST complete statement of its gas activity?

- (A) it only photosynthesises during the day and does not respire
- (B) it respire and also photosynthesises, so it takes in carbon dioxide and gives out oxygen overall in strong light
- (C) it only respire during the day and does not photosynthesise
- (D) it neither respire nor photosynthesises while sealed in light

26. The same leaf box is now kept in complete darkness all night. By morning the air inside has:

- (A) more carbon dioxide and less oxygen, because only respiration occurs in the dark
- (B) more oxygen, because the plant photosynthesises at night
- (C) no change, because plants do nothing at night
- (D) more oxygen and less carbon dioxide, just as in the day

27. In which plant parts does gas exchange for respiration mainly take place in leaves, in woody stems, and in roots, respectively?

- (A) lenticels; stomata; gills
- (B) spiracles; stomata; alveoli
- (C) stomata; lenticels; air spaces between soil particles
- (D) stomata; alveoli; lenticels

28. Limewater is bubbled with air breathed OUT and turns milky much faster than limewater bubbled with ordinary room air. The fairest conclusion is:

- (A) exhaled air contains no oxygen at all
- (B) room air contains no carbon dioxide at all
- (C) exhaled air contains more carbon dioxide than ordinary room air
- (D) exhaled air contains more oxygen than room air

29. A burning candle is lowered into a jar of air exhaled by a student and into a jar of fresh air. The candle goes out sooner in the exhaled-air jar. This is because exhaled air has:

- (A) less oxygen and more carbon dioxide than fresh air
- (B) more oxygen, which burns the candle out faster
- (C) the same gases as fresh air, so the result is chance
- (D) more water vapour, which is what put out the flame

30. Soaked germinating peas are sealed in a flask with a little limewater beside them for a day. The limewater turns milky and a candle lowered in afterward goes out. This shows the peas were:

- (A) respiring — using oxygen and giving out carbon dioxide
- (B) photosynthesising — giving out oxygen in the dark flask
- (C) dead, since seeds cannot respire
- (D) giving out only water vapour and no gas

31. Which product is shared by BOTH the aerobic respiration of glucose AND the anaerobic respiration of glucose by yeast?

- (A) lactic acid
- (B) oxygen
- (C) water as the only product
- (D) carbon dioxide

32. Which is the single TRUE statement about respiration?

- (A) only animals respire; plants merely photosynthesise
- (B) every living cell of both plants and animals respire to release energy
- (C) respiration stops in animals when they fall asleep
- (D) in a human only the lung cells respire

33. A sleeping baby keeps breathing steadily even though it is barely moving. The reason is that:

- (A) cells stop respiring in sleep, so air is stored for the morning
- (B) the lungs must make glucose at night and need air for it
- (C) breathing is only needed during movement and exercise
- (D) all its body cells keep respiring and need a continuous oxygen supply, even during rest

34. Which comparison of breathing across animals is CORRECT?

- (A) a cockroach uses gills while a fish uses spiracles
- (B) an earthworm uses lungs while a frog uses only gills
- (C) a frog uses spiracles while a fish uses moist skin
- (D) a fish uses gills while a cockroach uses tracheae fed by spiracles

35. A doctor presses a stethoscope to the chest and hears air moving. The structure just below the lungs that drives this air movement by changing the chest volume is the:

- (A) larynx
- (B) bronchus
- (C) diaphragm
- (D) alveolus

36. The hair and sticky mucus lining the nasal cavity matter for respiration because they:

- (A) trap dust and germs so cleaner air reaches the lungs
- (B) add extra oxygen to the air as it passes
- (C) release energy from the air for the body
- (D) turn carbon dioxide back into oxygen

37. A candle and a respiring mouse sealed in identical jars both eventually go quiet — the flame dies and the mouse becomes distressed. The common reason is that both:

- (A) used up the oxygen and built up carbon dioxide in the sealed jar
- (B) released oxygen and used up the carbon dioxide
- (C) needed sunlight that the jar blocked out
- (D) produced glucose until the jar was full

38. Put the inhalation events in the correct cause-and-effect order: (i) air flows into the lungs (ii) chest volume increases (iii) diaphragm flattens and ribs rise (iv) pressure inside the lungs falls.

- (A) i → iii → ii → iv
- (B) iii → ii → iv → i
- (C) iv → iii → ii → i
- (D) ii → i → iv → iii

39. A student writes 'breathing = respiration' in a test. The single best correction is:

- (A) breathing is the chemical release of energy; respiration is the exchange of gases
- (B) breathing is the mechanical exchange of gases; respiration is the chemical release of energy in cells
- (C) both are the same chemical reaction happening in the lungs
- (D) breathing happens in cells while respiration happens in the lungs

40. After fermentation in a flask, a lighted splint held over the gas above the liquid is snuffed out. This best identifies the gas as:

- (A) oxygen, which let the splint burn brighter
- (B) carbon dioxide, which does not support burning
- (C) hydrogen, which exploded with a pop
- (D) nitrogen, which the yeast breathed out

41. Which pairing of organism with its breathing organ is INCORRECT?

- (A) fish — gills
- (B) cockroach — tracheae
- (C) frog — gills only
- (D) human — lungs

42. A girl holds her breath, then has to breathe out forcefully. What mainly builds up in her blood and triggers the urge to breathe?

- (A) extra oxygen that the cells could not use
- (B) carbon dioxide produced by her respiring cells
- (C) lactic acid made by her lungs
- (D) alcohol from her muscles respiring

43. Why does a deep cut to the windpipe (trachea) make breathing difficult, even if the lungs themselves are healthy?

- (A) the trachea is the main airway, so damaging it blocks air from reaching the lungs
- (B) the trachea is where energy is released, so the cells starve
- (C) the trachea makes the oxygen that the lungs cannot
- (D) the trachea pumps blood, so breathing stops without it

44. A diagram labels a balloon-like structure where the air sac touches a fine blood vessel and oxygen crosses into the blood. This structure is the:

- (A) bronchus
- (B) alveolus
- (C) trachea
- (D) diaphragm

45. Two flasks of sugar solution: flask P has yeast and a balloon over its mouth; flask Q has yeast but is left open. After a day, P's balloon is inflated. The balloon inflated because:

- (A) carbon dioxide from the yeast's anaerobic respiration was trapped under the balloon
- (B) oxygen the yeast made collected under the balloon
- (C) the sugar solution boiled and the steam filled the balloon
- (D) air from the room leaked in and blew up the balloon

46. Compared with the air a person breathes IN, the air they breathe OUT has:

- (A) more oxygen, less carbon dioxide, and less water vapour
- (B) no oxygen at all and only carbon dioxide
- (C) less oxygen, more carbon dioxide, and more water vapour
- (D) exactly the same composition as the air breathed in

47. A swimmer takes one big breath and dives. Within seconds her muscles are still working hard underwater. Which respiration is increasingly likely in those muscles as the dive continues?

- (A) anaerobic respiration, because oxygen runs short and lactic acid begins to form
- (B) photosynthesis, because the muscles make their own oxygen
- (C) aerobic respiration with no change, because one breath lasts the whole dive
- (D) no respiration at all, to save oxygen

48. A respirometer shows a small animal using up oxygen steadily in a sealed chamber over an hour. This steady oxygen use is direct evidence that the animal is:

- (A) photosynthesising, since it gives out oxygen
- (B) respiring continuously to release energy
- (C) only breathing, with no respiration happening inside
- (D) storing the oxygen unused in its body

49. Both a fish and a human ultimately need oxygen for the same purpose. That shared purpose is to:

- (A) make food by photosynthesis using the oxygen
- (B) produce glucose in their breathing organs
- (C) turn carbon dioxide directly into energy
- (D) release energy from food inside their cells by respiration

50. Which observation would best convince a doubter that PLANTS respire, not only animals?

- (A) a leaf in bright sunlight gives off bubbles of oxygen
- (B) a sealed jar of germinating seeds in the dark turns limewater milky and snuffs a candle
- (C) a plant grows taller over several weeks in the garden
- (D) a flower opens in the morning and closes at night

51. During quiet exhalation, which describes the ribs, diaphragm, and lung pressure correctly?

- (A) ribs move up and out, diaphragm flattens, and lung pressure falls
- (B) ribs stay still, diaphragm flattens, and lung pressure falls
- (C) ribs move down and in, diaphragm domes up, and lung pressure rises
- (D) ribs move down and in, diaphragm flattens, and lung pressure falls

52. A person's breathing rate is 16/min at rest and rises to 24/min while walking quickly. The percentage idea this best illustrates is that exercise:

- (A) lowers the breathing rate to save oxygen during effort
- (B) keeps the breathing rate fixed regardless of effort
- (C) stops breathing temporarily while the muscles work
- (D) raises the breathing rate above the resting value to meet higher oxygen demand

53. Why does food (glucose) need oxygen brought by breathing before its energy can be fully released?

- (A) oxygen turns the glucose into more glucose first
- (B) oxygen cools the cell so the glucose does not burn away
- (C) oxygen is the energy itself, stored inside each cell
- (D) oxygen lets the glucose be completely broken down to carbon dioxide and water, releasing the most energy

54. A wind instrument player blows hard for a long phrase and then gasps. The gasping afterward mainly serves to:

- (A) release stored alcohol made by the lungs
- (B) take in oxygen quickly and remove the carbon dioxide that built up while blowing
- (C) cool the air inside the windpipe
- (D) make glucose to refill the muscles

55. A model 'bicycle pump' is sometimes compared to the lungs. The fair part of the comparison is that both:

- (A) release energy by breaking down glucose
- (B) extract oxygen from water like gills
- (C) change the volume of a space to move air in and out
- (D) turn limewater milky on their own

56. Which is the correct full word equation for aerobic respiration as taught in Class 7?

- (A) glucose + carbon dioxide → oxygen + water + energy
- (B) carbon dioxide + water + energy → glucose + oxygen
- (C) glucose + oxygen → carbon dioxide + water + energy
- (D) glucose → carbon dioxide + water + energy

57. A sealed jar with a burning candle, a second sealed jar with a respiring grasshopper, and a third sealed jar with germinating seeds are each left for an hour. Limewater in all three jars turns milky. The single idea this demonstrates is that:

- (A) burning and respiration in animals and plant seeds all release carbon dioxide
- (B) only the candle releases carbon dioxide; the living things absorb it
- (C) the seeds release oxygen while the others release carbon dioxide
- (D) carbon dioxide is only produced when something is on fire

58. An athlete finishes a hard race and keeps breathing deeply and fast for several minutes even after stopping. The best explanation is that:

- (A) extra oxygen is still needed to break down the lactic acid that built up during the race
- (B) the lungs are stuck open and cannot slow down at once
- (C) the body is releasing the alcohol made by the leg muscles
- (D) breathing must stay fast to keep making lactic acid

59. A student lists structures: trachea, bronchi, alveoli, nasal cavity. Ordered from where incoming air arrives FIRST to LAST, the correct order is:

- (A) trachea → nasal cavity → bronchi → alveoli
- (B) nasal cavity → bronchi → trachea → alveoli
- (C) nasal cavity → trachea → bronchi → alveoli
- (D) alveoli → bronchi → trachea → nasal cavity

60. On a cold morning a person sees 'mist' when they breathe out, but not when they breathe in. This is direct evidence that exhaled air contains more:

- (A) oxygen, which freezes into a white cloud
- (B) carbon dioxide, which is visible as a white gas
- (C) nitrogen, which the cold turns into mist
- (D) water vapour, which condenses into tiny droplets in the cold air

Solutions — Science (Class 7), Chapter 10: Respiration in Organisms — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	B	21	A	31	D	41	C	51	C
2	D	12	A	22	D	32	B	42	B	52	D
3	D	13	D	23	D	33	D	43	A	53	D
4	A	14	B	24	B	34	D	44	B	54	B
5	D	15	C	25	B	35	C	45	A	55	C
6	C	16	B	26	A	36	A	46	C	56	C
7	C	17	C	27	C	37	A	47	A	57	A
8	A	18	D	28	C	38	B	48	B	58	A
9	C	19	C	29	A	39	B	49	D	59	C
10	B	20	D	30	A	40	B	50	B	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) The equation produces carbon dioxide and water (both collectable materials) plus energy. Energy is what the cell uses to do work; it is not matter, so it cannot be collected in a jar. Water is genuinely a product, so saying it is not produced is wrong. Carbon dioxide is produced, not used up. Oxygen is on the left (a reactant), not the right, so that option misreads the equation.

2. (D) Breathing is the gaseous exchange that brings oxygen to the blood; respiration is the energy release inside cells that consumes that oxygen. When muscles work hard, they respire faster and need more oxygen, so breathing speeds up to supply it — the two are linked, not unrelated. Faster breathing does not slow respiration. Respiration occurs in every cell, not just the lungs.

3. (D) In aerobic respiration the glucose is fully broken down to carbon dioxide and water, releasing a large amount of energy. In anaerobic respiration glucose is only partly broken down (to alcohol or lactic acid), so much energy stays locked in those products and far less

is released. Anaerobic respiration uses no oxygen, so it cannot 'use extra oxygen.' Alcohol is an anaerobic (yeast) product, not aerobic. Speed is not the reason for the lower yield; incomplete breakdown is.

4. (A) Yeast in dough has little oxygen, so it respire anaerobically: glucose $\rightarrow$ alcohol + carbon dioxide + energy. The carbon dioxide gas forms bubbles that expand the dough and make it rise. Yeast does not photosynthesise (it is not green and there is no light in dough), so it does not give off oxygen. The dough is not hot enough to boil into vapour. Lactic acid is a muscle product, not a yeast product.

5. (D) Yeast breaks down sugar without oxygen, producing alcohol (the smell) and carbon dioxide (the bubbles) — the basis of brewing. The plain juice has no yeast, so neither change occurs there, meaning it did not respire faster. Yeast does not run the reaction backwards into sugar and oxygen, and it does not photosynthesise.

6. (C) During the sprint the muscle ran short of oxygen and respired anaerobically, building up lactic acid that causes cramp. On resting, breathing continues to deliver oxygen, which clears the lactic acid, so the cramp eases. Rest does not make MORE lactic acid. Muscles make lactic acid, not alcohol (that is yeast). Refilling with carbon dioxide would not relieve a cramp.

7. (C) Anaerobic respiration in muscles happens during sudden, heavy exertion when oxygen supply cannot keep up with demand, so lactic acid forms. A sudden heavy lift fits this. Reading, gentle walking, and sleeping are all low-demand activities where oxygen supply easily meets demand, so respiration stays aerobic.

8. (A) The lungs carry out breathing — they take in oxygen and give out carbon dioxide. Energy is released by respiration, which happens inside every living cell, not in the lungs. So 'lungs make energy' is wrong even during exercise. The nose only lets air in and cleans it. Blood carries gases but does not 'make energy' as it passes the lungs.

9. (C) The increase is $28 - 16 = 12$ breaths per minute. 28 is the jogging rate itself, not the increase. 44 is the two rates added ($28 + 16$). 16 is the resting rate. The question asks for the difference, which is 12.

10. (B) Half an hour is 30 minutes, so $15 \times 30 = 450$ breaths. 45 would be only three minutes (15×3). 900 would be a full hour (15×60). 30 is just the number of minutes, not breaths. The correct product is 450.

11. (B) The numbers climb from sleeping (12) up to running (36) as activity increases, showing breathing rate rises with effort because more oxygen is needed and more carbon dioxide must be removed. The data clearly rise, not fall, so 'falls as activity increases' is wrong. They change, so 'stays the same' is wrong. The lowest rate is during sleep (least activity), not the highest, so 'fastest at rest' is wrong.

12. (A) A trained body uses oxygen more efficiently, so the athlete can do the same work with fewer breaths — A's lower rate signals better fitness, not worse. A higher rate (B) means the body is working harder to keep up, so B is not the fitter one. A low rate does not mean respiration has stopped, and all living cells need oxygen, so 'A's cells do not need oxygen' is false.

13. (D) High up, the air is thinner (less dense), so each breath delivers less oxygen; the body breathes faster to make up the shortfall. The cause is the thin air, not the cold acting on the diaphragm. Mountain air is not 'full of carbon dioxide.' The lungs do not physically shrink with altitude; the oxygen content of each breath is what drops.

14. (B) Air enters the nostrils, passes the nasal cavity, goes down the trachea (windpipe), splits into the bronchi, reaches the alveoli, and oxygen crosses into the blood there. The other orders jumble the sequence — putting the nasal cavity after the trachea, or the nostrils after the nasal cavity, or the bronchi before the trachea. Only nostrils → nasal cavity → trachea → bronchi → alveoli → blood is anatomically correct.

15. (C) Inhalation: the diaphragm contracts, flattening and moving down, while the ribs swing up and out — together these enlarge the chest cavity so air is drawn in. The other options mix a downward diaphragm with inward ribs, or a domed diaphragm with outward ribs, which contradict each other or describe exhalation. Only flatten-down diaphragm plus up-and-out ribs gives a larger chest.

16. (B) When the chest cavity enlarges, the lungs expand and the air pressure inside them drops below atmospheric pressure; air then flows from high to low pressure, into the lungs. The lungs have no muscle to actively 'suck' — they are passive. The diaphragm enlarges the cavity but does not push air up from the stomach. Temperature differences are not what drive the airflow; the pressure difference does.

17. (C) On exhaling, the diaphragm relaxes (domes up) and the ribs move down and in, so the chest shrinks; this raises the pressure inside the lungs above the outside air, and air flows out. An expanding chest with falling pressure describes inhalation, not exhalation. The lungs do not manufacture air to push out, and the ribs move down and in (not further up and out) during exhalation.

18. (D) The rubber sheet at the bottom acts like the diaphragm. Pulling it down enlarges the space in the bottle, lowering the pressure so the balloon (lung) inflates — exactly what happens in inhalation when the diaphragm flattens downward. Pulling down cannot model exhalation, when the diaphragm domes up. The sheet is the diaphragm, not the ribs; the balloon, not the sheet, is the lung.

19. (C) Insects breathe through spiracles (the small openings) that lead into a branching network of tracheae carrying air straight to the tissues. Gills and bronchi belong to fish and

humans. Stomata and lenticels are plant structures. Nostrils and alveoli are parts of the human system. Only spiracles and tracheae describe a cockroach.

20. (D) Fish gills extract oxygen dissolved in water. Boiling expels the dissolved oxygen, so the gills find little to absorb and the fish struggles — this is a shortage, not an excess, of oxygen. Fish do not breathe carbon dioxide. The trouble is the missing oxygen, not the water's purity or the fish 'swallowing' it.

21. (A) A given volume of water contains far less oxygen than the same volume of air, so a fish must move a large amount of water across its gills to get enough oxygen — hence the near-constant pumping. It is not that fish 'respire many times faster' as cells; it is the low oxygen content of water. Gills can take oxygen from still water too (movement just helps); the real limit is how little oxygen water holds.

22. (D) Earthworms take in oxygen through their moist skin from air in the soil spaces. After heavy rain, water fills those spaces, leaving too little air, so worms come up where oxygen is available. They do not 'drink water to breathe.' Earthworms have no gills. Waterlogged soil has LESS available air-oxygen, not extra, so they are escaping a shortage, not an excess.

23. (D) An adult frog breathes through both its lungs (in air) and its moist skin, and the skin can absorb oxygen dissolved in water, letting it stay submerged. Adult frogs do not have fish-like gills. It does not store oxygen in its stomach. A living frog cannot stop respiring — its cells always need energy — so 'stops respiring completely' is false.

24. (B) A fish uses gills, a cockroach uses spiracles, and an earthworm uses its moist skin — all correct. The second set swaps gills and spiracles and gives the earthworm lungs, all wrong. The third gives a fish moist skin and an earthworm gills, both wrong. The fourth gives a fish lungs and an earthworm spiracles, both wrong.

25. (B) Plants respire all the time, day and night. In bright daylight they also photosynthesise, usually faster than they respire, so the net exchange is taking in carbon dioxide and giving out oxygen. It is wrong to say a plant does not respire by day — respiration never stops. It clearly does photosynthesise in light, and it is active (not idle) while sealed.

26. (A) In darkness there is no photosynthesis, but respiration continues, so the plant takes in oxygen and gives out carbon dioxide all night — the box ends up with more CO₂ and less O₂. Plants do not photosynthesise without light, so it does not gain oxygen at night. They are not idle; respiration goes on. The day-time net result (more O₂) needs light, which is absent here.

27. (C) Leaves exchange gases through stomata, woody stems through lenticels, and roots obtain oxygen from air in the spaces between soil particles. The first wrong set swaps stomata and lenticels and adds gills (an animal organ). The others bring in spiracles and

alveoli, which are animal structures, not plant ones. Only stomata / lenticels / soil air spaces is correct for a plant.

28. (C) Limewater turns milky in proportion to carbon dioxide, so faster milkiness with exhaled air shows exhaled air is richer in CO₂ than room air. It does not show exhaled air has zero oxygen — exhaled air still contains plenty of oxygen, just less than fresh air. Room air does contain a small amount of CO₂ (it just turns limewater milky slowly). Exhaled air has LESS oxygen, not more.

29. (A) A flame needs oxygen. Exhaled air has less oxygen (used up by respiration) and more carbon dioxide, so the candle goes out sooner. More oxygen would help a flame, not snuff it. The result is consistent and not chance, since the gases differ. The extra water vapour is real but the main reason the flame dies is the reduced oxygen, not the moisture.

30. (A) Milky limewater shows carbon dioxide was produced and the candle going out shows oxygen was used up — both signs of respiration in the living, germinating seeds. They cannot be photosynthesising to make oxygen in a flask with no light. Living seeds clearly do respire, so 'dead' is wrong. Carbon dioxide (not just water vapour) was released, proven by the limewater.

31. (D) Aerobic respiration gives carbon dioxide + water; yeast's anaerobic respiration gives alcohol + carbon dioxide. The common product is carbon dioxide. Lactic acid comes from muscle anaerobic respiration, not yeast or aerobic. Oxygen is a reactant in aerobic respiration, not a product. Water is an aerobic product but is not the 'only' product, and it is not produced by yeast fermentation.

32. (B) Respiration is a property of all living cells, plant and animal, because all need energy. Plants both photosynthesise and respire, so 'only animals respire' is false. Respiration never stops while alive — even a sleeping animal's cells keep respiring. And in a human every cell respire, not just lung cells (the lungs only handle gas exchange).

33. (D) Cells never stop respiring while an organism is alive, so oxygen must be supplied continuously, even in sleep — hence steady breathing. Cells do not pause respiration to 'store air.' Lungs do not make glucose. Breathing is needed at all times, not only during exercise, because resting cells still need energy.

34. (D) A fish breathes through gills; a cockroach takes air in through spiracles into tracheae — both correct. The first wrong option swaps gills and spiracles between fish and cockroach. The second gives the earthworm lungs (it uses skin) and the frog only gills (an adult frog uses skin and lungs). The third gives the frog spiracles and the fish moist skin, both wrong.

35. (C) The diaphragm is the dome-shaped muscle below the lungs; its movement changes the chest volume to draw air in and push it out. The larynx (voice box) sits above

the windpipe and does not drive breathing. A bronchus is an airway branch, not a pump. An alveolus is a tiny air sac for gas exchange, not a muscle.

36. (A) Nasal hair and mucus clean the incoming air by trapping dust and germs before it reaches the delicate lungs. They cannot add oxygen — the air's oxygen comes from outside. They do not release energy; that happens inside cells during respiration. They certainly cannot convert carbon dioxide into oxygen.

37. (A) Both a flame and a respiring animal consume oxygen and release carbon dioxide; in a sealed jar the oxygen runs out and CO₂ builds up, so the flame dies and the mouse struggles. Neither releases oxygen or uses up CO₂ — that is the reverse. Sunlight is irrelevant to a candle and a mouse. Neither produces glucose; cells consume glucose.

38. (B) Inhalation begins with the muscles moving (diaphragm flattens, ribs rise), which increases chest volume, which lowers the lung pressure, which finally lets air flow in: iii → ii → iv → i. The other orders place air flowing in first, or pressure dropping before the muscles move, or volume changing before the muscles act — all reverse cause and effect.

39. (B) Breathing is a physical/mechanical process of moving air in and out (gas exchange); respiration is the chemical process inside cells that releases energy from glucose. The reversed option swaps the two definitions. They are not the same reaction, and respiration is not confined to the lungs. The last option reverses where each occurs.

40. (B) Fermentation (yeast respiring without oxygen) gives off carbon dioxide, which does not support burning, so it puts the splint out. Oxygen would make the splint burn brighter, not go out. Hydrogen would give a 'pop,' not a quiet snuffing, and is not a fermentation product. Yeast does not breathe out nitrogen; the snuffing gas is CO₂.

41. (C) An adult frog breathes through its moist skin and lungs, not gills only, so that pairing is the incorrect one the question wants. A fish does use gills, a cockroach does use tracheae (via spiracles), and a human does use lungs — all correct pairings. So the false one is the frog with 'gills only.'

42. (B) Even while holding the breath, cells keep respiring and producing carbon dioxide, which builds up in the blood and drives the urge to breathe. Oxygen is being used up, not building up. Lungs do not make lactic acid (muscles do, and not from simple breath-holding). Muscles make lactic acid, not alcohol; alcohol is a yeast product.

43. (A) The trachea (windpipe) is the main tube carrying air to the bronchi and lungs; if it is damaged, air cannot reach healthy lungs, so breathing is impaired. Energy is released in cells, not in the trachea. The trachea does not 'make' oxygen — air carries it. The trachea is an airway, not a blood pump.

44. (B) Oxygen passes into the blood at the alveoli — thin-walled sacs in close contact with blood capillaries. A bronchus is a larger branching airway, not where gas crosses into

blood. The trachea is the windpipe leading down to the bronchi. The diaphragm is the breathing muscle, not a gas-exchange surface.

45. (A) Yeast respire the sugar anaerobically, releasing carbon dioxide; in the sealed flask P this gas is trapped and inflates the balloon. Yeast does not make oxygen. The solution does not boil at room temperature, so it is not steam. Room air did not leak in — the gas was produced inside by the yeast, which is why the open flask Q does not inflate a balloon.

46. (C) Respiration uses oxygen and produces carbon dioxide and water, so exhaled air has less oxygen, more carbon dioxide, and more water vapour than inhaled air. It is not richer in oxygen. It still contains plenty of oxygen (just less), so 'no oxygen at all' is wrong. And the composition clearly changes, so 'exactly the same' is wrong.

47. (A) As the working muscles use up the limited oxygen during the dive, they increasingly switch to anaerobic respiration, forming lactic acid. Muscles cannot photosynthesise to make oxygen. One breath cannot keep hard-working muscles fully aerobic indefinitely. Respiration cannot stop in living muscle — cells always need energy.

48. (B) Steady consumption of oxygen shows the animal's cells are respiring continuously, using oxygen to release energy. Using up oxygen is the opposite of photosynthesis (which gives out oxygen), so that is wrong. Breathing supplies oxygen precisely so respiration can use it, so 'no respiration inside' makes no sense. The oxygen is being used, not stored unused.

49. (D) Whatever organ takes oxygen in — gills for the fish, lungs for the human — the oxygen is ultimately used by cells to release energy from food through respiration. Neither animal photosynthesises. Glucose comes from food, not from the breathing organs. Energy is released from glucose using oxygen; carbon dioxide is a waste product, not a source of energy.

50. (B) Respiration produces carbon dioxide (milky limewater) and uses oxygen (candle goes out); seeds in the dark show this because there is no photosynthesis to mask it — proof that plants respire. Oxygen bubbles in sunlight show photosynthesis, not respiration. Growth and flowers opening do not directly demonstrate gas exchange of respiration.

51. (C) In exhalation the ribs drop down and in and the diaphragm relaxes upward (domes up), shrinking the chest and raising the lung pressure so air is pushed out. Ribs up/diaphragm flat/pressure falling describes inhalation. A flattening diaphragm enlarges the chest (inhalation), so any option pairing a flat diaphragm with exhalation is wrong.

52. (D) The rate climbs from 16 to 24 with exercise, which shows effort raises breathing rate to supply more oxygen and clear more carbon dioxide. The data rise, so exercise does not lower or fix the rate, and breathing certainly does not stop during effort — it speeds up.

53. (D) Oxygen allows the glucose to be broken down completely into carbon dioxide and water, which releases the full amount of energy (aerobic respiration). Oxygen does not multiply glucose. Its role is chemical breakdown, not cooling. Oxygen is a gas used in the reaction, not 'energy itself' — the energy comes from breaking down the glucose.

54. (B) While holding and pushing out a long phrase, the player's intake of oxygen is reduced and carbon dioxide accumulates, so the gasp afterward restores oxygen and clears carbon dioxide. Lungs do not make alcohol. The gasp is about gas exchange, not cooling the windpipe. Glucose is not made by breathing — it comes from food.

55. (C) A pump and the chest both work by changing a volume to move air — the valid comparison. A pump does not respire or break down glucose. It does not extract oxygen from water (that is a gill's job). A pump does not produce carbon dioxide to turn limewater milky on its own; that comes from respiration.

56. (C) Aerobic respiration: $\text{glucose} + \text{oxygen} \rightarrow \text{carbon dioxide} + \text{water} + \text{energy}$. The version with carbon dioxide as a reactant and oxygen as a product is photosynthesis-like and wrong. The reversed equation is photosynthesis. Leaving out oxygen on the left describes anaerobic respiration, not aerobic. Only $\text{glucose} + \text{oxygen} \rightarrow \text{carbon dioxide} + \text{water} + \text{energy}$ is correct for aerobic respiration.

57. (A) All three turn limewater milky, showing carbon dioxide is released by the candle (burning), the grasshopper (animal respiration), and the seeds (plant respiration). The living things release CO₂, not absorb it. The seeds respire and give out CO₂ (not oxygen) in the jar. CO₂ is clearly produced without fire too, by the living respiring organisms.

58. (A) During the race the muscles respired anaerobically and built up lactic acid; afterwards the body needs extra oxygen to break that lactic acid down, so deep fast breathing continues for a while. The lungs are not mechanically 'stuck.' Muscles make lactic acid, not alcohol. Breathing continues to REMOVE the lactic acid, not to make more of it.

59. (C) Air entering reaches the nasal cavity first, then the trachea, then branches into the bronchi, and finally the alveoli where gas exchange occurs. Putting the trachea before the bronchi, or the bronchi before the trachea, breaks the real route. The last option is the reverse (exit) order. Only nasal cavity → trachea → bronchi → alveoli is correct for incoming air.

60. (D) Respiration produces water, so exhaled air is rich in water vapour; in cold air this vapour condenses into visible mist. Oxygen does not freeze into a cloud at that temperature. Carbon dioxide is a colourless gas and is not what you see. Nitrogen does not condense into mist in ordinary cold air. The visible mist is condensing water vapour.